and also in a right angle triangle OPM, $\angle OPM = \theta_2$ which implies, $\varphi_2 = \frac{\pi}{2} + \theta_2$

Hence,

$$\vec{B} = \frac{\mu_\circ I}{4\pi a}\left(\cos\left(\frac{\pi}{2} - \theta_1\right) - \cos\left(\frac{\pi}{2} + \theta_2\right)\right)\hat{n}$$

$$= \frac{\mu_\circ I}{4\pi a}(\sin\theta_1 + \sin\theta_2)\hat{n}$$

3.8.3 Magnetic field produced along the axis of the current carrying circular coil

Consider a current carrying circular loop of radius R and let I be the current flowing through the wire in the direction as shown in Figure 3.40. The magnetic field at a point P on the axis of the circular coil at a distance z from its center of the coil O. It is computed by taking two diametrically opposite line elements of the coil each of length $d\vec{l}$ at C and D. Let $\vec{r}$ be the vector joining the current element (I $d\vec{l}$) at C to the point P.

$$PC = PD = r = \sqrt{R^2 + Z^2} \text{ and}$$
$$\text{angle } \angle CPO = \angle DPO = \theta$$

According to Biot-Savart's law, the magnetic field at P due to the current element $I\ d\vec{l}$ is

$$d\vec{B} = \frac{\mu_\circ}{4\pi}\frac{I\,d\vec{l} \times \hat{r}}{r^2}$$

The magnitude of magnetic field due to current element $I\ d\vec{l}$ at C and D are equal because of equal distance from the coil. The magnetic field $d\vec{B}$ due to each current element $I\ d\vec{l}$ is resolved into two components; dB sin θ along y - direction and dB cos θ along z - direction. Horizontal components of each current element cancels out while the vertical components (dB cos θ $\hat{k}$) alone contribute to total magnetic field at the point P.

If we integrate $d\vec{l}$ around the loop, $d\vec{B}$ sweeps out a cone as shown in Figure 3.40, then the net magnetic field $\vec{B}$ at point P is

$$\vec{B} = \int d\vec{B} = \int dB\cos\theta\ \hat{k}$$

$$\vec{B} = \frac{\mu_\circ I}{4\pi}\int \frac{dl}{r^2}\cos\theta\ \hat{k}$$

But $\cos\theta = \frac{R}{\left(R^2 + Z^2\right)^{\frac{1}{2}}}$, using Pythagorous theorem $r^2 = R^2 + Z^2$ and integrating line element from 0 to 2πR, we get

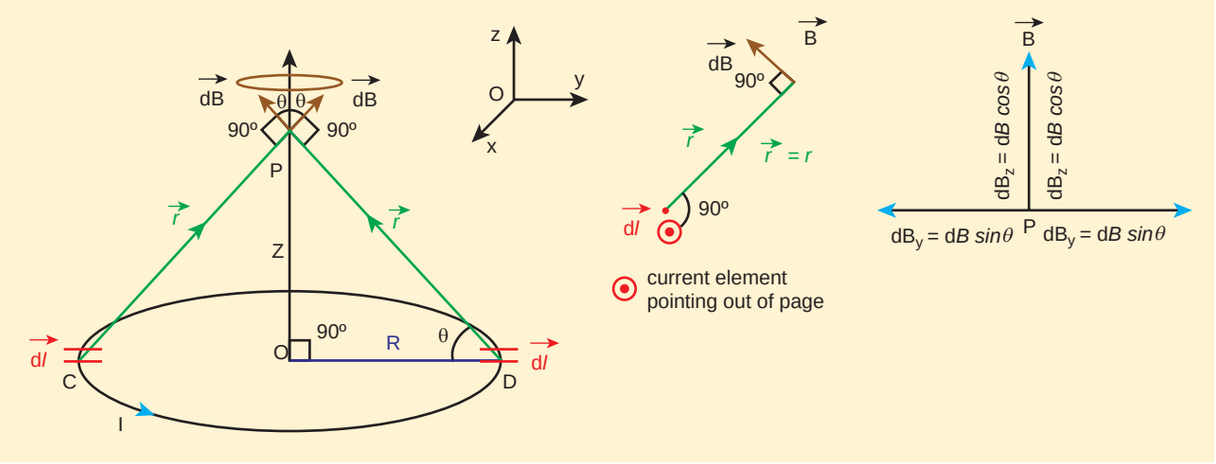


Figure 3.40 Current carrying circular loop using Biot-Savart's law

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